

Flow Level Anomaly Detection under Domain Shift

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Abstract

Anomaly detectors trained on one capture site degrade when deployed on another. We quantify that degradation and test whether a shared encoder narrows the gap.

1 Introduction

We study anomaly detectors that must survive a change of capture site. Operational telemetry differs between sites in packet rate, protocol mix, and the prevalence of tunnelling, so a detector tuned on one capture rarely keeps its operating point on another.

2 Approach

The encoder maps each flow record to a dense vector of fixed width. Vectors are compared with cosine distance against a reference population drawn from the deployment site itself, which removes the need for labelled attacks at deployment time.

2.1 Encoder

The encoder is a six layer residual network over bucketed packet descriptors, trained with a contrastive objective on unlabelled flows from both sites.

3 Evaluation

We report area under the ROC curve and recall at a fixed alert budget for both capture sites, using the same threshold selection procedure throughout.

Table 1: Detection quality per capture site.

Site	AUC	Recall
Site A	0.971	0.884
Site B	0.902	0.771

Site B is the harder deployment: its benign population is broader and its alert budget is smaller.

3.1 Ablation

Depth helps until it does not: doubling the encoder doubles parameters for a modest gain, which matters when the detector runs on a capture appliance.

Table 2: Ablation over encoder depth.

Depth	AUC	Params
6	0.944	21M
12	0.971	42M

We therefore keep the six layer encoder for all deployment experiments.

4 Discussion

A shared encoder narrows but does not close the cross site gap. The remaining difference tracks the prevalence of tunnelled traffic, which suggests a data problem rather than a modelling one.

References

- [1] F. Observer. Domain shift in network telemetry. Example Letters, 2021.
- [2] G. Practitioner. Deployment notes for flow detectors. Example Reports, 2023.